

OLS statistical model fit results

Model definitions

Single image:

$$Y \sim \text{const} + \beta(a)*\text{age} + \beta(s)*\text{sex} + \beta(g)*\text{group} + \beta(r)*\text{religion} + \beta(i)*\text{group}*\text{religion}$$

Pair of images:

$$Y(2) - Y(1) \sim \text{const} + \beta(a)*\text{age} + \beta(s)*\text{sex} + \beta(g)*\text{group} + \beta(r)*\text{religion} + \beta(i)*\text{group}*\text{religion}$$

Where:

Y: STD of Hilbert envelope averaged over [1,2] s interval
in specified frequency band;

$\beta(\dots)$ regression coefficient (“slope”, “correlation”)

age: as is;

sex: as is;

group: 0 for SCZ, 1 for ASD;

religion: Santa-Clara score

Values of interest:

Regression coefficients (“slopes”, “correlations”) for **group**,
religion and **group*religion variables**

Interpretation

Model:

$$Y \sim \text{const} + \dots + \beta(g)*\text{group} + \beta(r)*\text{religion} + \beta(i)*\text{group}*\text{religion}$$

“group” main effect $\beta(g)$: describes difference in induced power between ASD and SCZ groups given religion score = 0

“religion” main effect $\beta(r)$: is interpreted as a correlation “beta” between induced power and SC score **for group = 0** (SCZ):

$$Y = \text{const} + \beta(r)*\text{religion}; \quad \text{group} = 0$$

“interaction” effect $\beta(i)$: describes a **difference in induced power/SC score correlation** between ASD and SCZ groups.

“Religion” slope for ASD group: plug in group = 1 into the model equation:

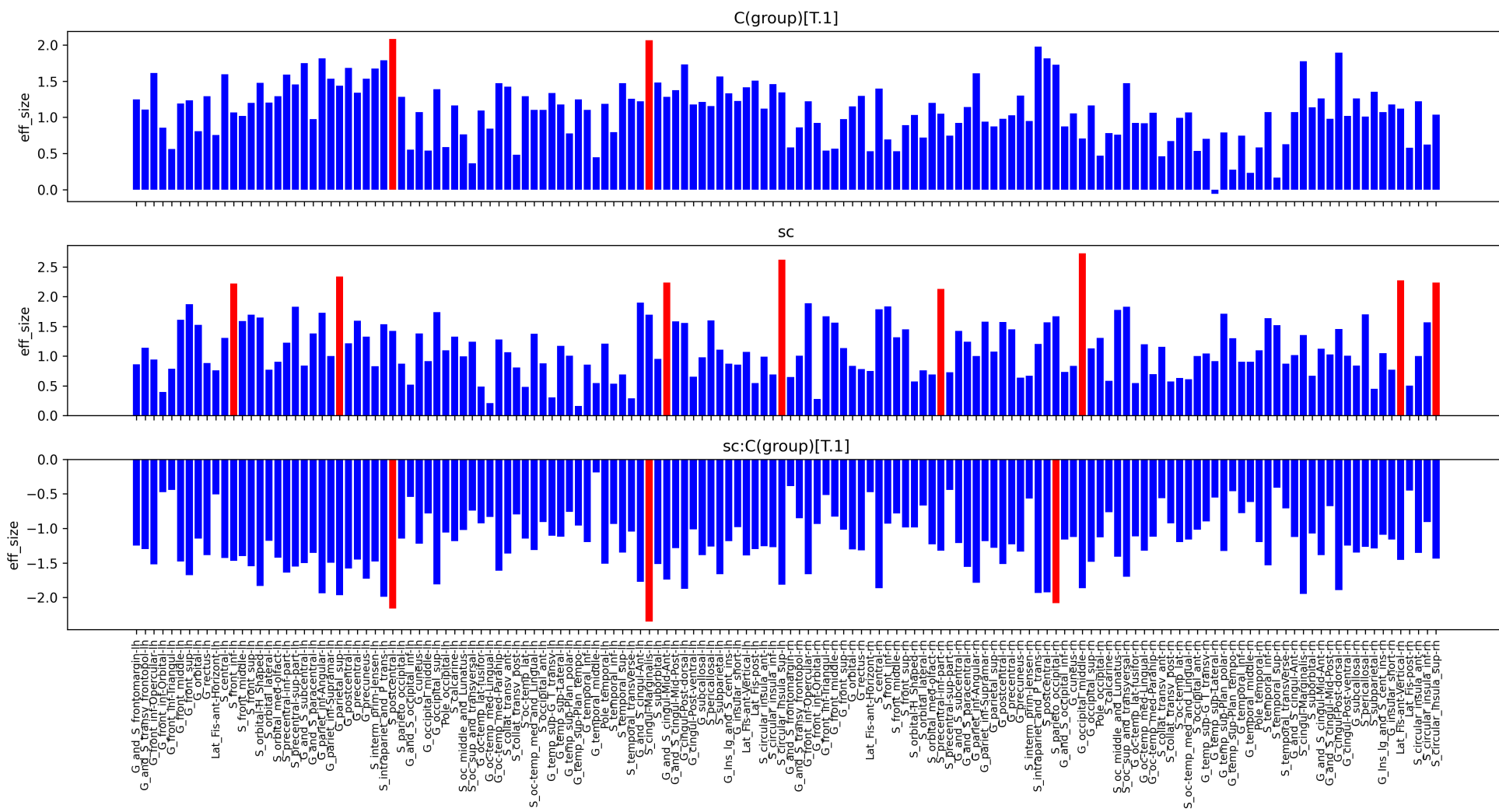
$$Y = \text{const} + \beta(r)*\text{religion} + \beta(i)*\text{religion}; \quad \text{group} = 1$$

Thus **“religion” slope for ASD group** is:

$$\beta(r \text{ for ASD}) = \beta(r) + \beta(i)$$

4 – 7 Hz, image = GOD

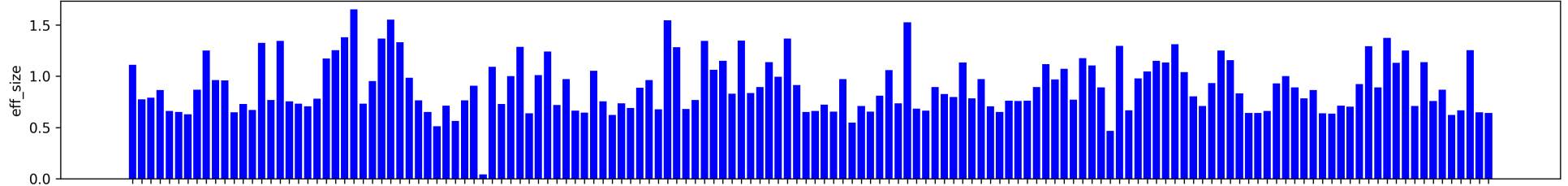
Task: 'henv std sm santaimg', [4.0, 7.0] Hz, img = God



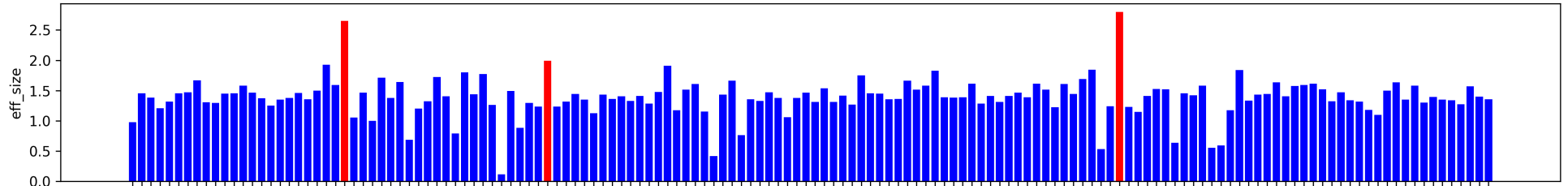
4 – 7 Hz, image = THING

Task: 'henv_std_sm_santa1img', [4.0, 7.0] Hz, img = Thing

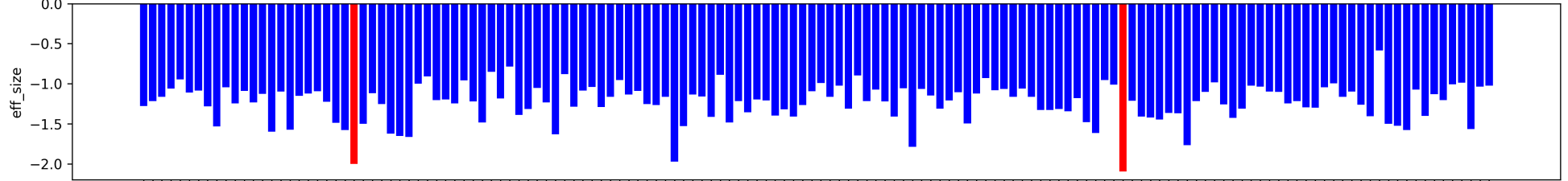
C(group)[T.1]



SC



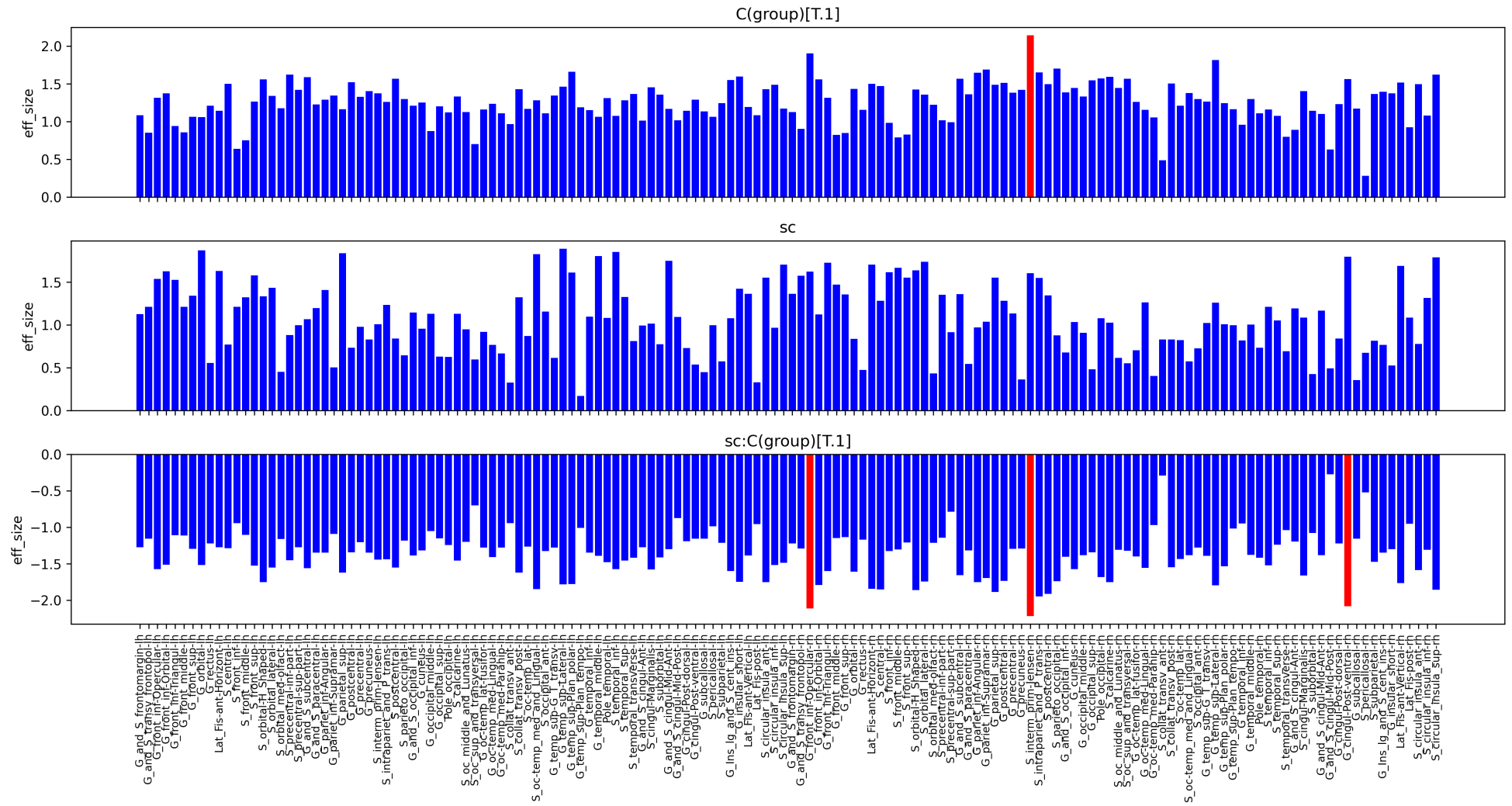
sc:C(group)[T.1]



G_and_S_frontomarginal
G_and_S_transverse
G_front_inf-Opercular
G_front_inf-Orbital
G_front_inf-Triangular
G_front_middle
G_orbita
G_orbita_inf
Lat_Fis-ant-Horizont
S_central
S_central_inf
S_front_middle
S_orbita
S_orbita_inf
S_orbita_med-Occipital
S_precentral_inf-part
S_precentral_inf
G_and_S_subcentral
G_and_S_subcentral_inf
G_pariet_inf-Supramarginal
G_parietal_sup
G_postcentral
G_precuneus
S_interm_pfm-densen
S_intrapariet_and_P_intrapariet
S_parieto_occipital
G_and_S_occipital_inf
G_occipital_middle
G_occipital_sup
Pole-calcarine
S_oc_middle_and_Lunatus
S_oc_sup_and_transverse
G_oc-temp_med-Parahipp
G_oc-temp_med-Lingual
S_collat_transv_post
S_oc-temp_lat
S_oc-temp_med-Occipital
G_temp_sup-G_L_transv
G_temp_sup-Lateral
G_temp_sup-Plan-tempo
G_tempora_inf
G_tempora_middle
Pole-tempora
S_tempora_inf
S_tempora_sup
G_and_S_Marginalis
G_and_S_cingul_Ant
G_and_S_cingul_Mid-Ant
G_and_S_cingul_Mid-Post
G_cingul_Post-dorsal
G_cingul_Post-lateral
G_cingul_subcallosa
S_pericallosa
S_subcallosa
G_ins_lg_and_S_cent_ins
Lat_Fis-ant-Vertical
S_circular_insula_ant
S_circular_insula_inf
G_and_S_frontrorhinal
G_and_S_transv_frontrorhinal
G_front_inf-Orbital
G_front_inf-Triangular
G_front_middle
G_rectus
Lat_Fis-ant-Horizont
G_rectus_inf
S_front_middle
S_orbita_H_Shaped
S_orbita_lateral
S_precentral_inf-part
S_precentral_inf
G_and_S_subcentral
G_and_S_subcentral_inf
G_pariet_inf-Angular
G_pariet_inf-Supramarginal
G_postcentral
G_precuneus
S_interm_pfm-densen
S_intrapariet_and_P_intrapariet
S_parieto_occipital
G_and_S_occipital_inf
G_cuneus
G_occipital_middle
G_occipital_sup
Pole-occipital
S_calcarine
S_oc_middle_and_Lunatus
G_oc-sup_and_transverse
G_oc-temp_lat-fusiform
G_oc-temp_med-Occipital
G_oc-temp_med-Lingual
S_collat_transv_ant
S_collat_transv_post
G_and_S_cingul_Ant
G_and_S_cingul_Mid-Ant
G_and_S_cingul_Mid-Post
G_cingul_Post-dorsal
G_cingul_Post-lateral
G_subcallosa
S_subcallosa
G_ins_lg_and_S_cent_ins
Lat_Fis-ant-Vertical
S_circular_insula_ant
S_circular_insula_inf
S_circular_insula_sup

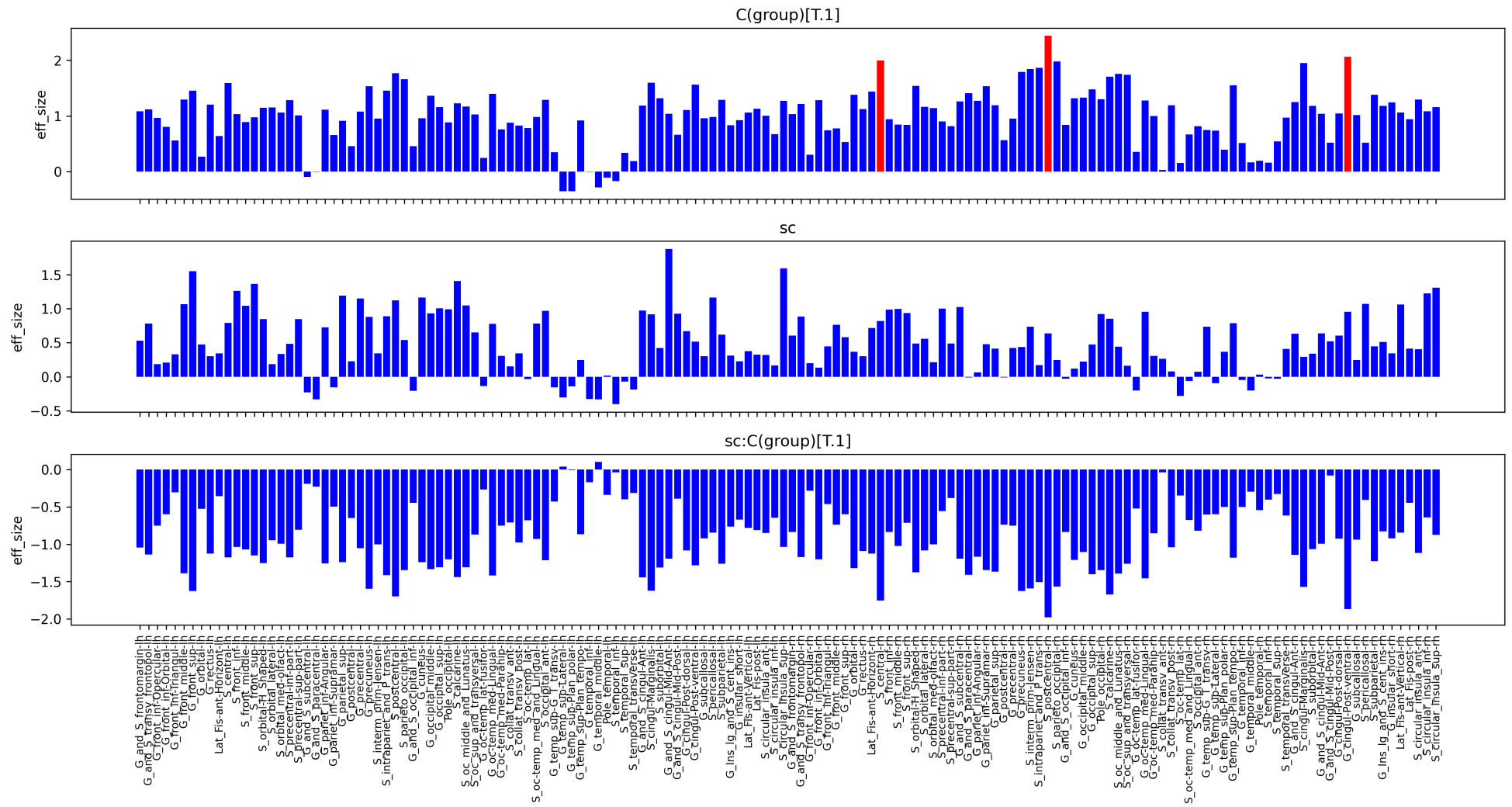
4 – 7 Hz, image = HUMAN

Task: 'henv_std_sm_santa1img', [4.0, 7.0] Hz, img = Human



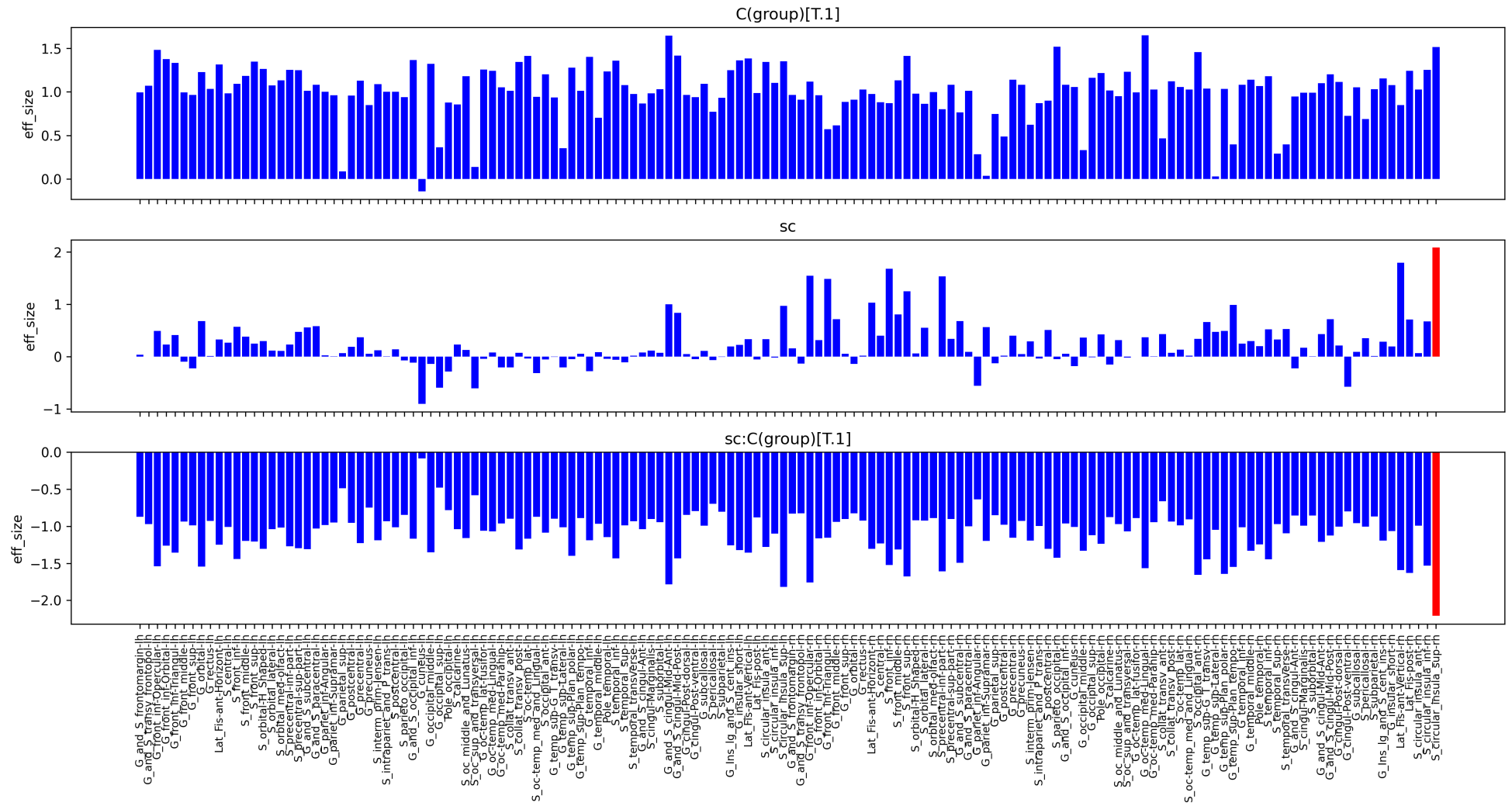
4 – 7 Hz, image = SUPER

Task: 'henv_std_sm_santa1img', [4.0, 7.0] Hz, img = Super



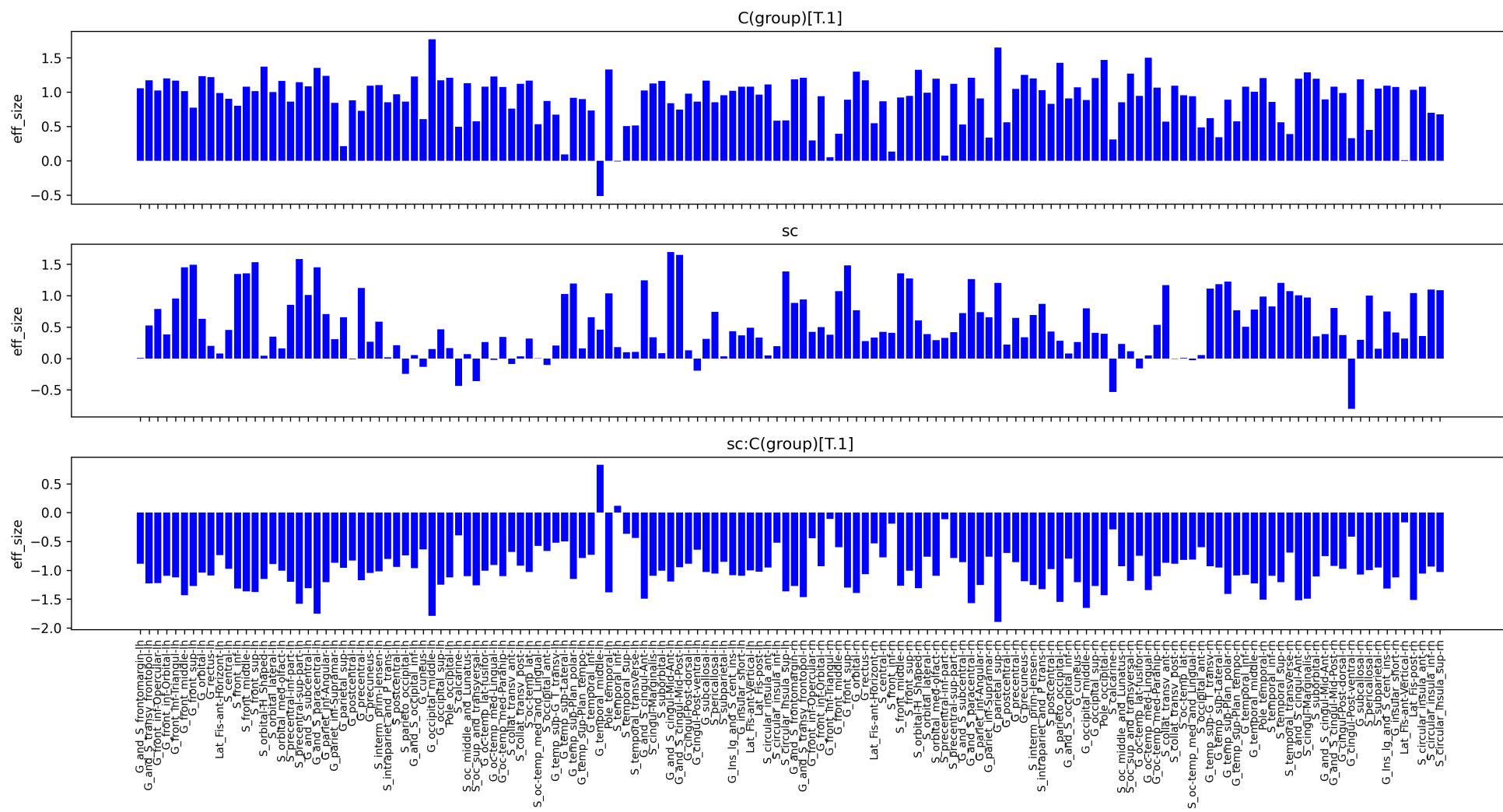
30 – 50 Hz, image = GOD

Task: 'henv_std_sm_santa1img', [30.0, 50.0] Hz, img = God



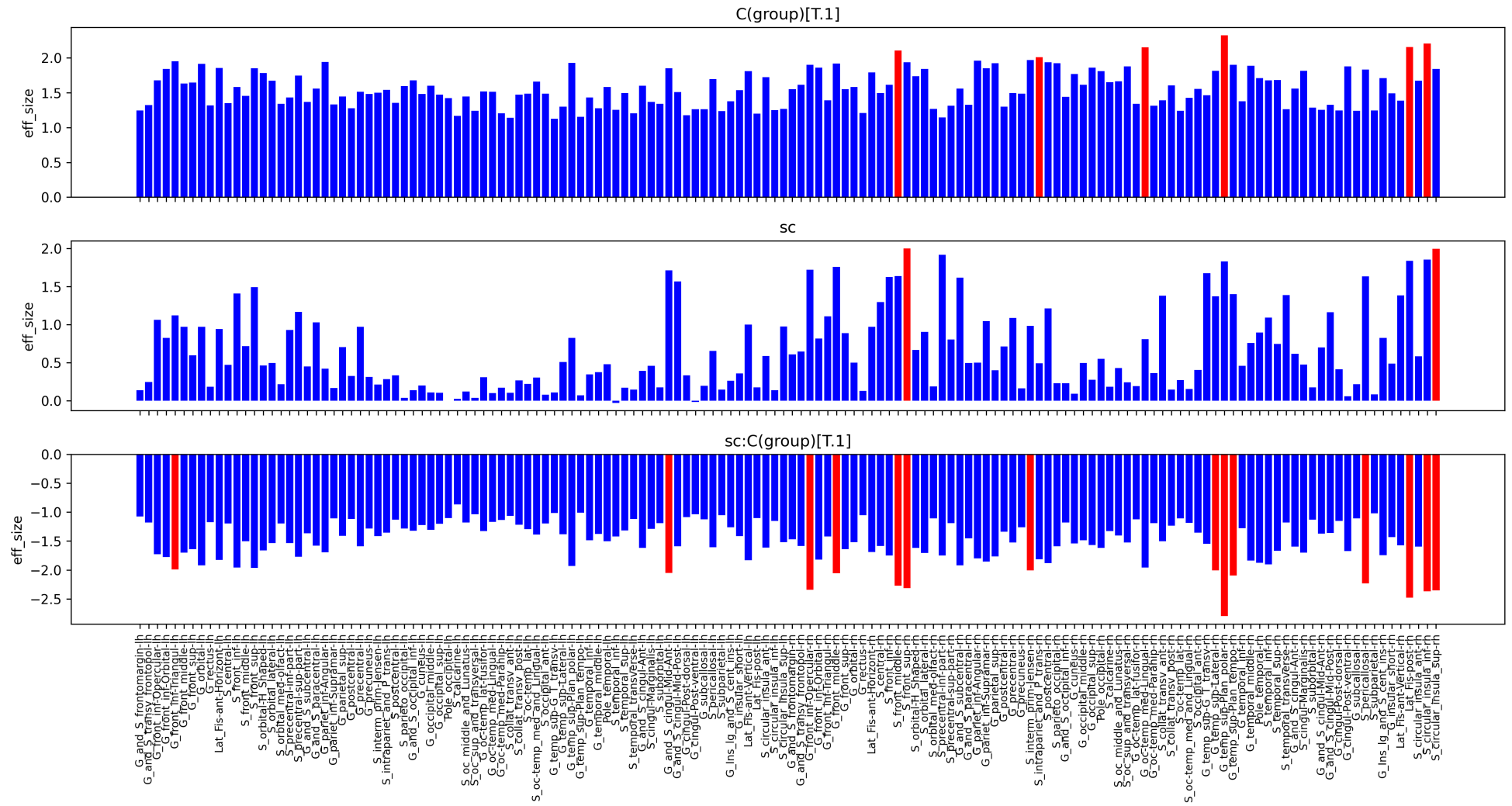
30 – 50 Hz, image = THING

Task: 'henv std sm santa1img', [30.0, 50.0] Hz, img = Thing



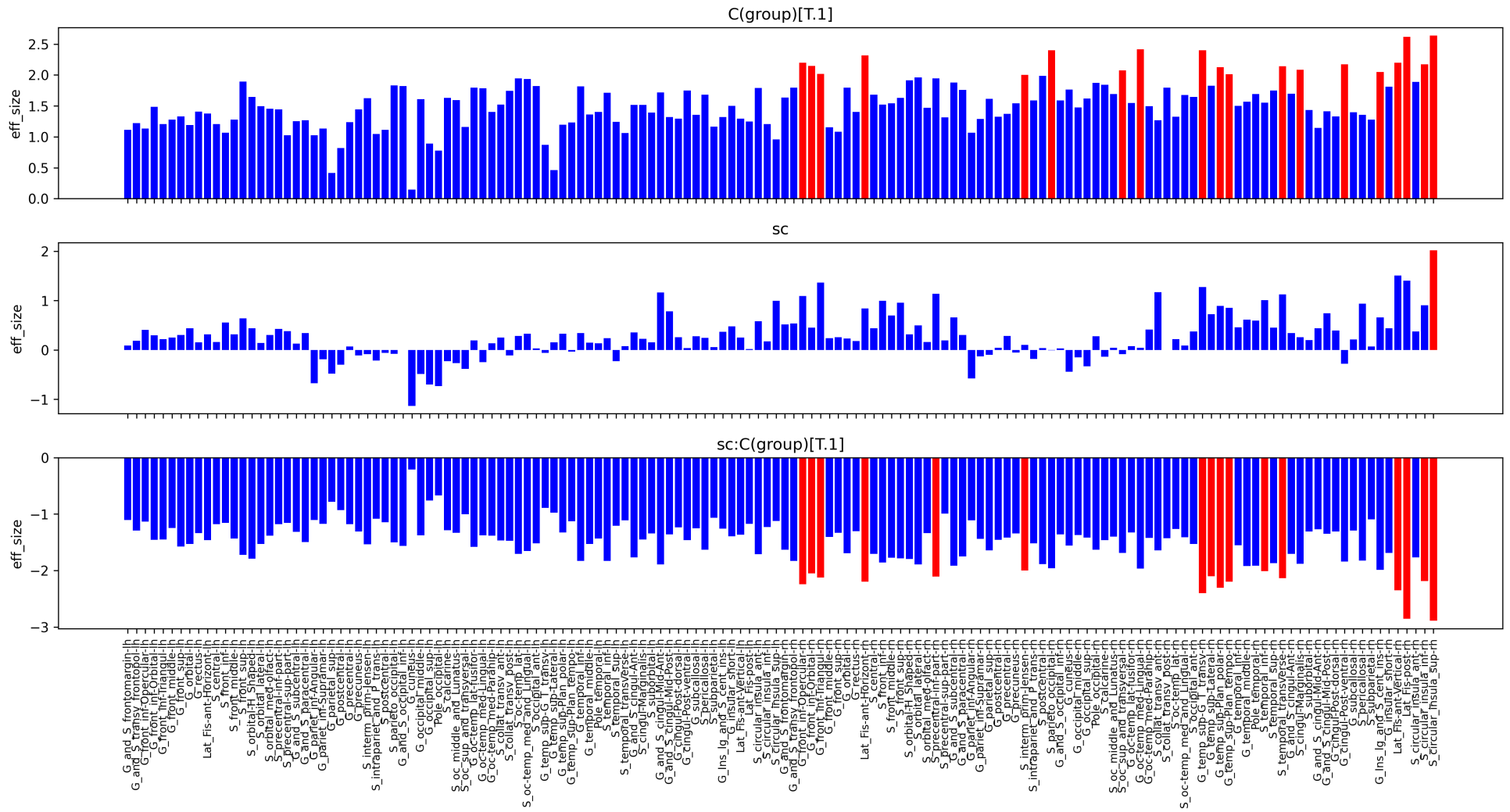
30 – 50 Hz, image = HUMAN

Task: 'henv_std_sm_santa1img', [30.0, 50.0] Hz, img = Human

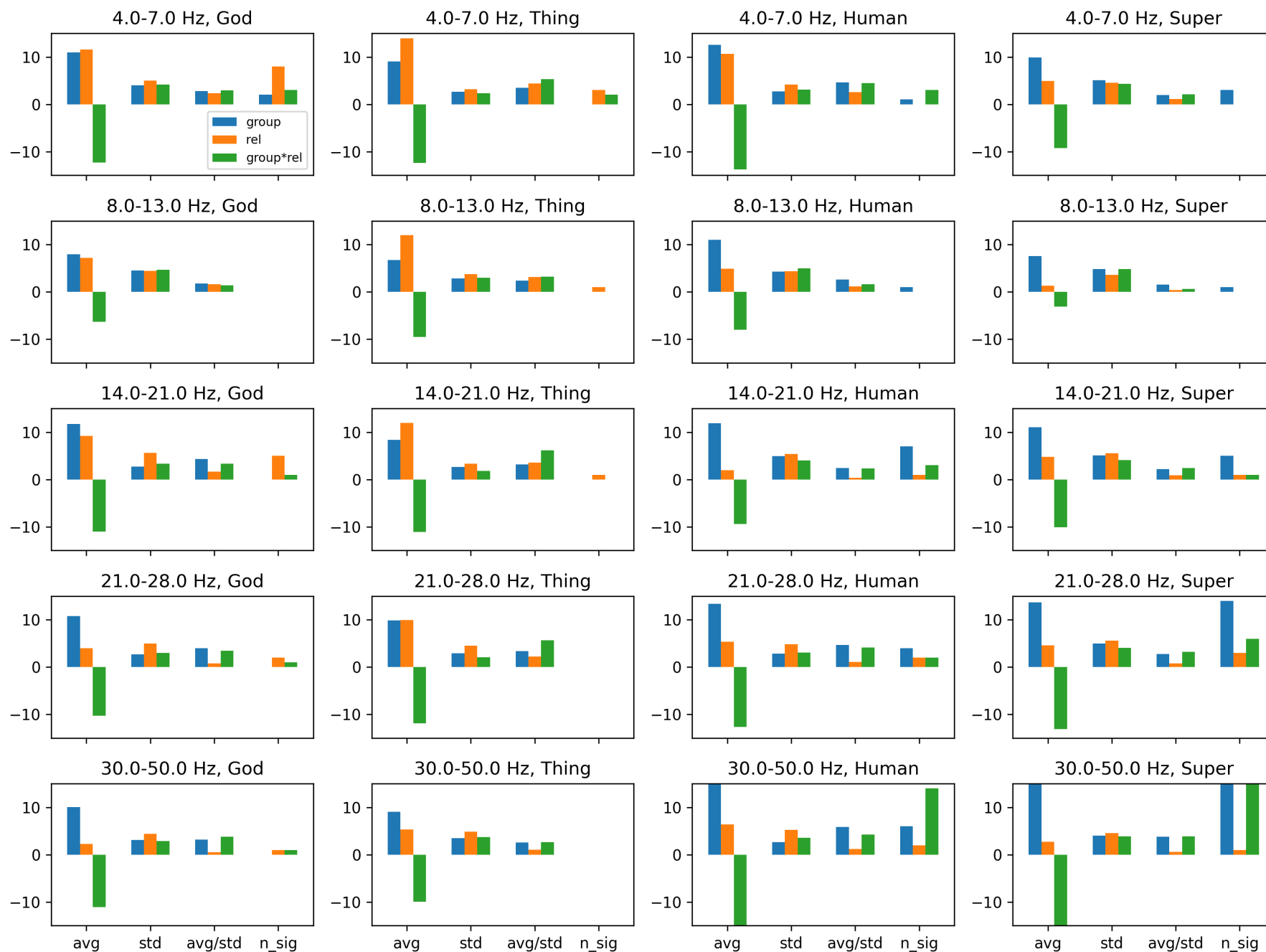


30 – 50 Hz, image = SUPER

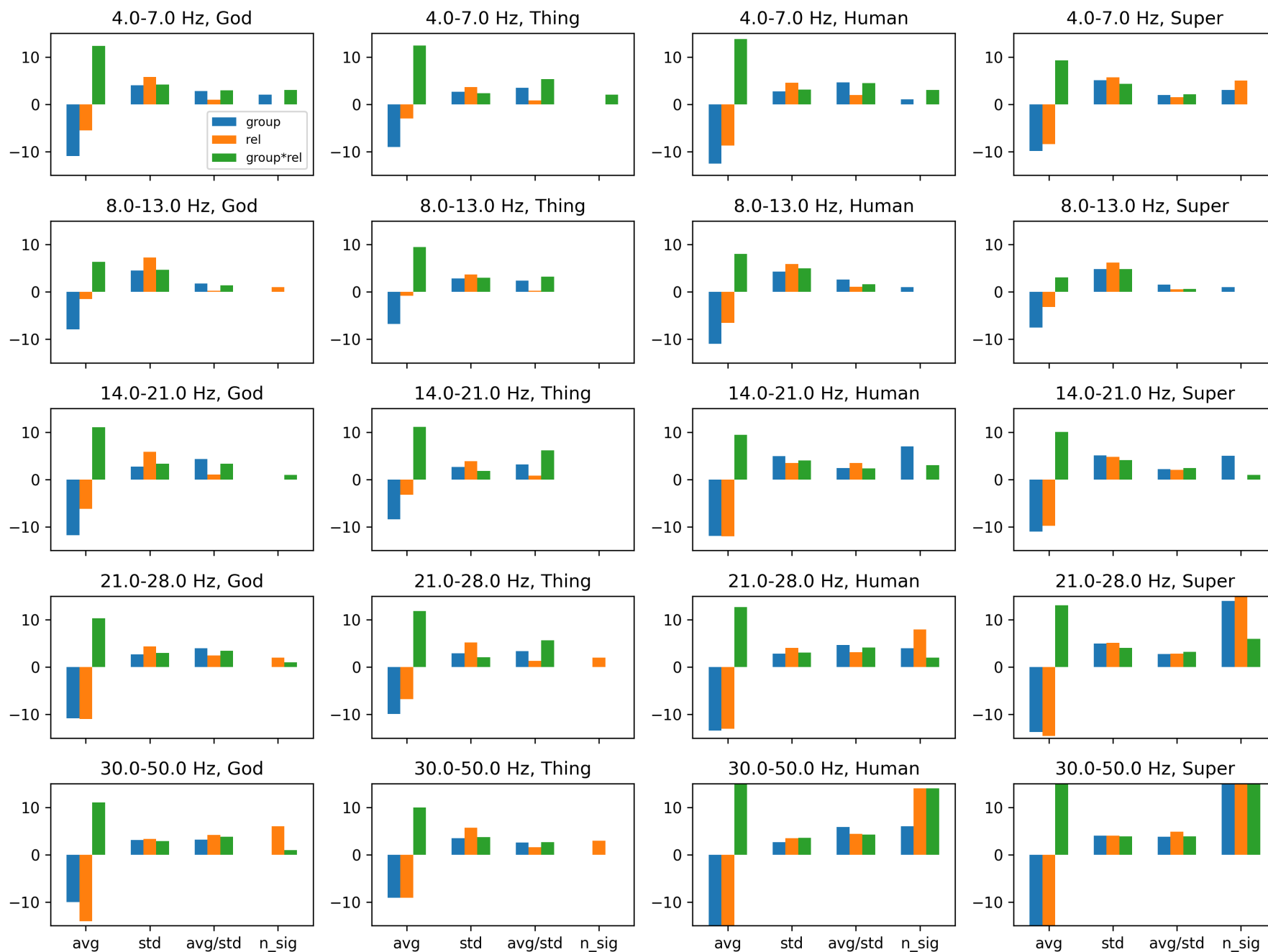
Task: 'henv_std_sm_santa1img', [30.0, 50.0] Hz, img = Super



Single image results overview



Single image results overview – groups swapped



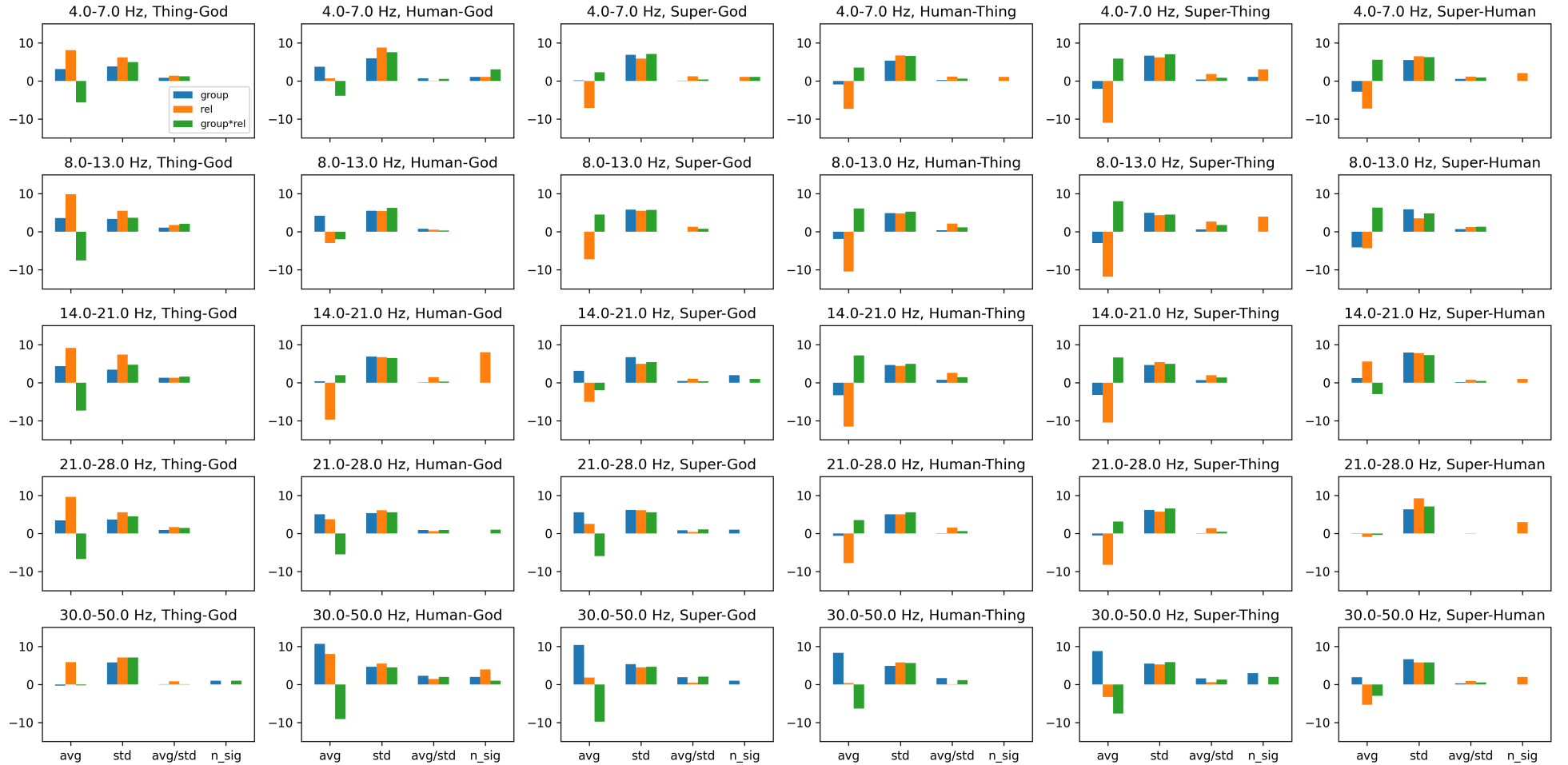
Summary for single image results

- Individual ROI results are **not statistically significant**
- For all bands and images **event-related power is larger for ASD**
- This group effect is typically smallest for the "Thing" image, and largest for "Human" image.
- The largest values are obtained for Human and Super for [21-28] and [30-50] frequency bands
- Religiosity effect is predominantly positive – that is **for SCZ group response power mostly increases with the SC score** irrespective of the image type
- Strong religiosity effect is seen in lower frequency bands, for God and Thing, however overall dependency on the image type varies
- Consistency of religiosity slope over ROIs is smaller than for the group effect. The most stable (over ROI) response to SC is for Thing. The lowest consistency is for higher frequencies.
- Interaction term is ****predominantly NEGATIVE****, meaning that slopes for ASD are always less than for SCZ group.

By swapping the group numbers, it can be shown that overall **religiosity effect for ASD group is predominantly negative**.

- Strongest religiosity effects are observed in higher frequency bands (21-28, 30-50) Hz and for image types Super and Thing, and in the right hemisphere.

Image pairs results overview



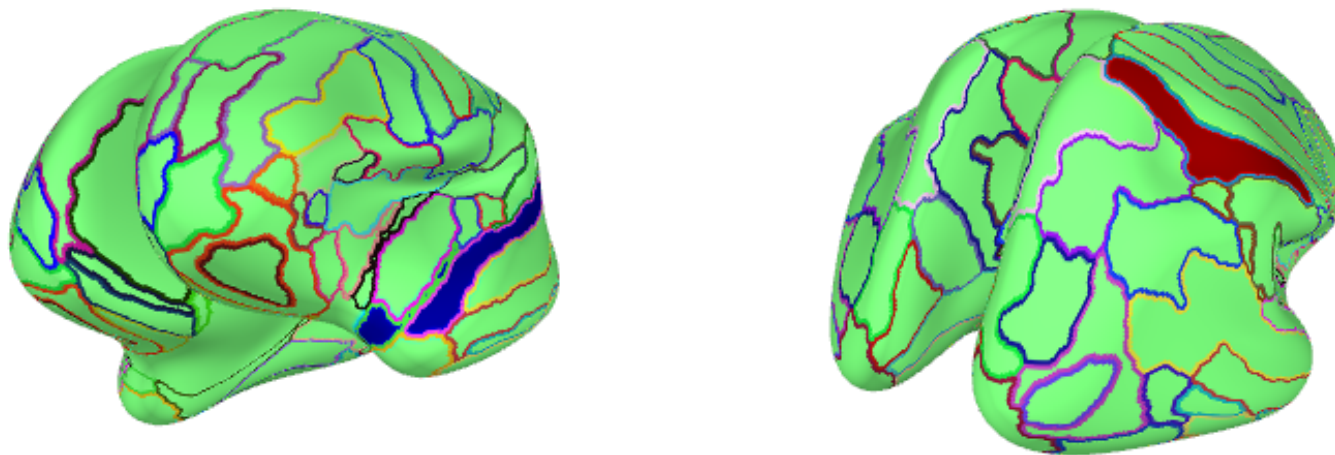
ROI-specific single image results

- As individual ROI results are not statistically significant, we chose to show those below and above 1% (2-sided) significant level.
- The significance thresholds were calculated individually for each fitted parameter, by pooling results from all frequency bands.
- The significance thresholds for effect sizes corresponding to 0.5% and 99.5% percentiles are shown in the table below.

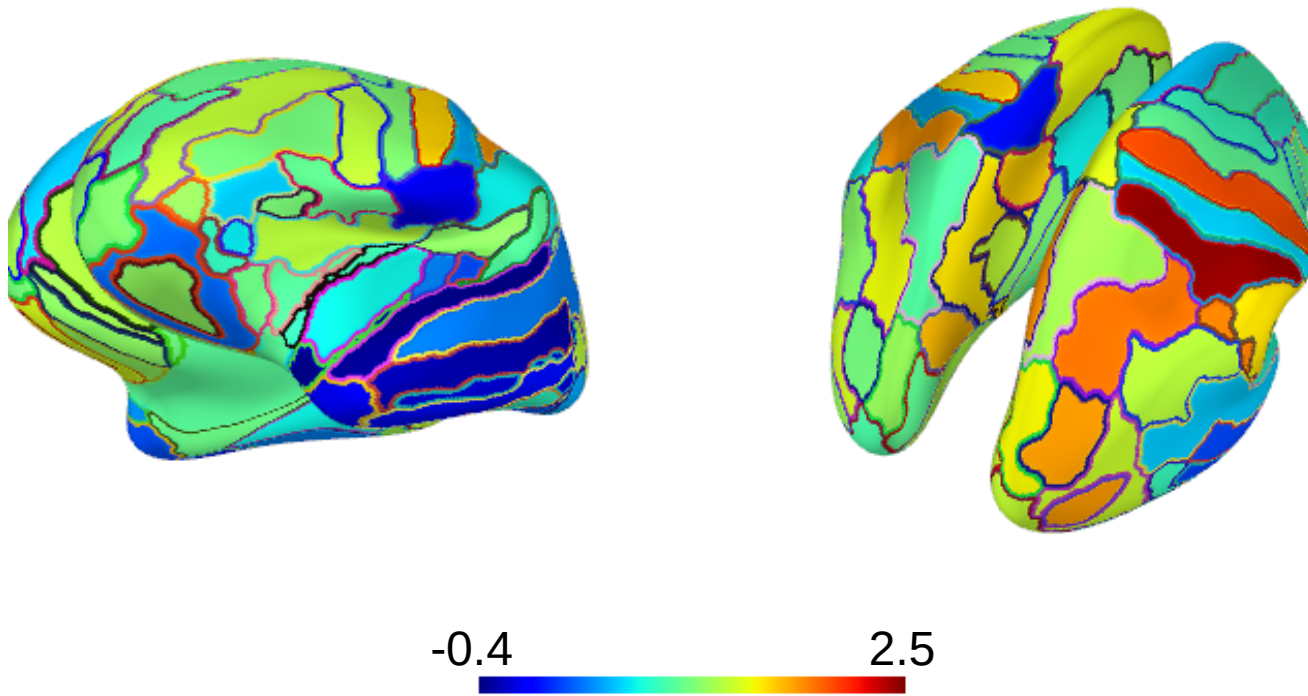
	Lower (0.5%)	Upper (99.5%)
group	-0.31	2.24
rel	-0.65	2.2
group*rel	-2.25	0.53

Table 1. 0.5% and 99.5% percentile thresholds for effect sizes

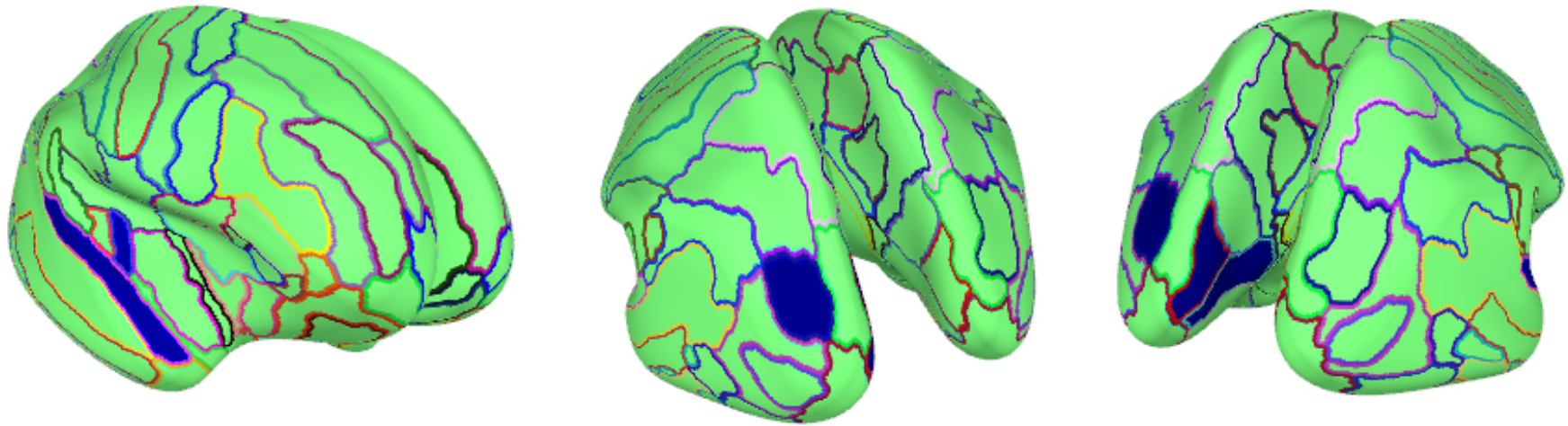
4 – 7 Hz, parm = group, img = “Super”, thresholded



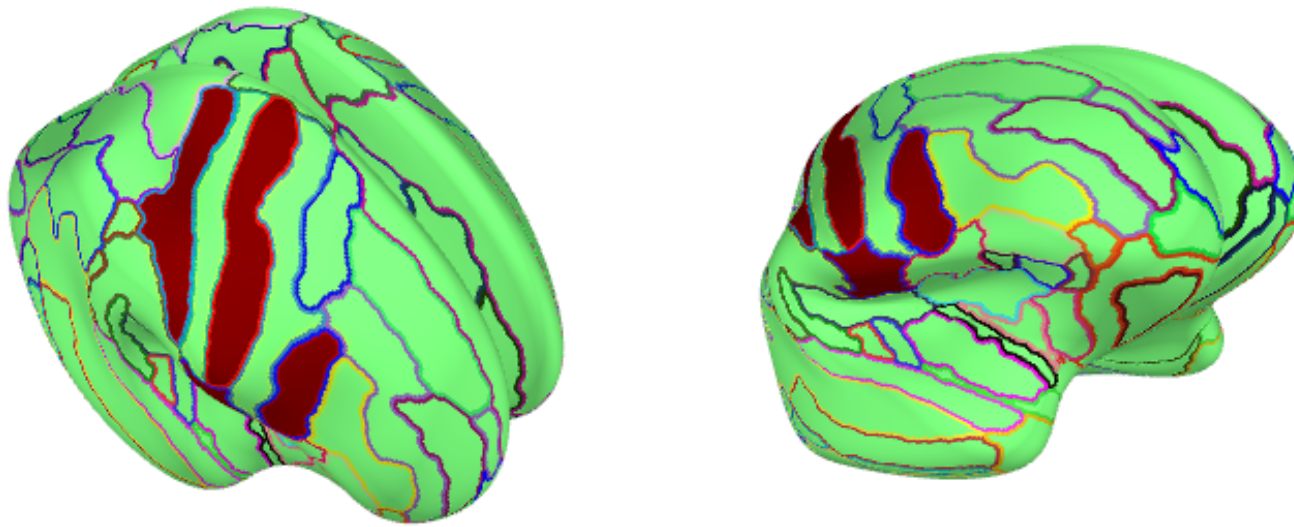
4 – 7 Hz, parm = group, img = “Super”, unthresholded



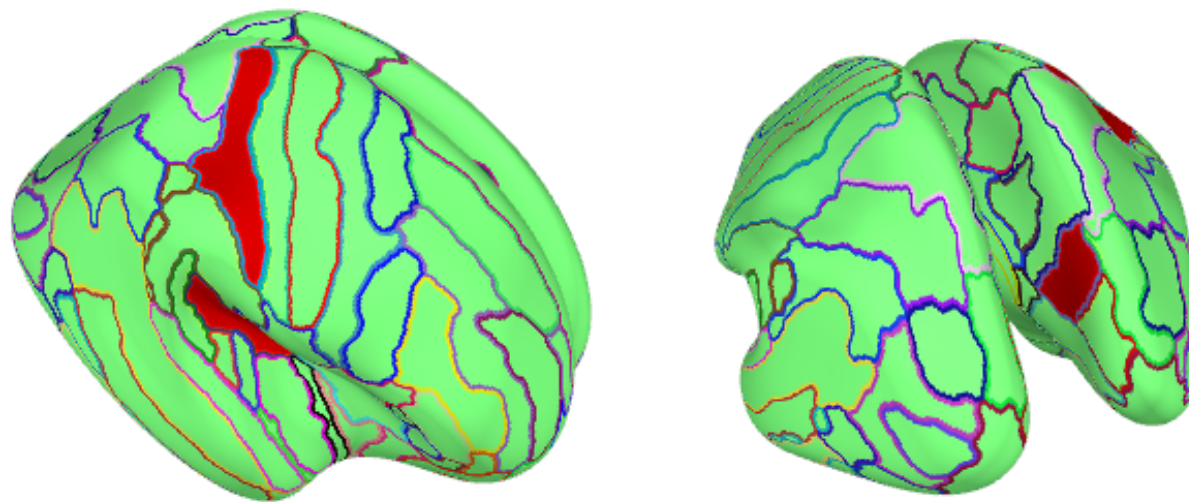
8 – 13 Hz, parm = group, img = “God”, thresholded



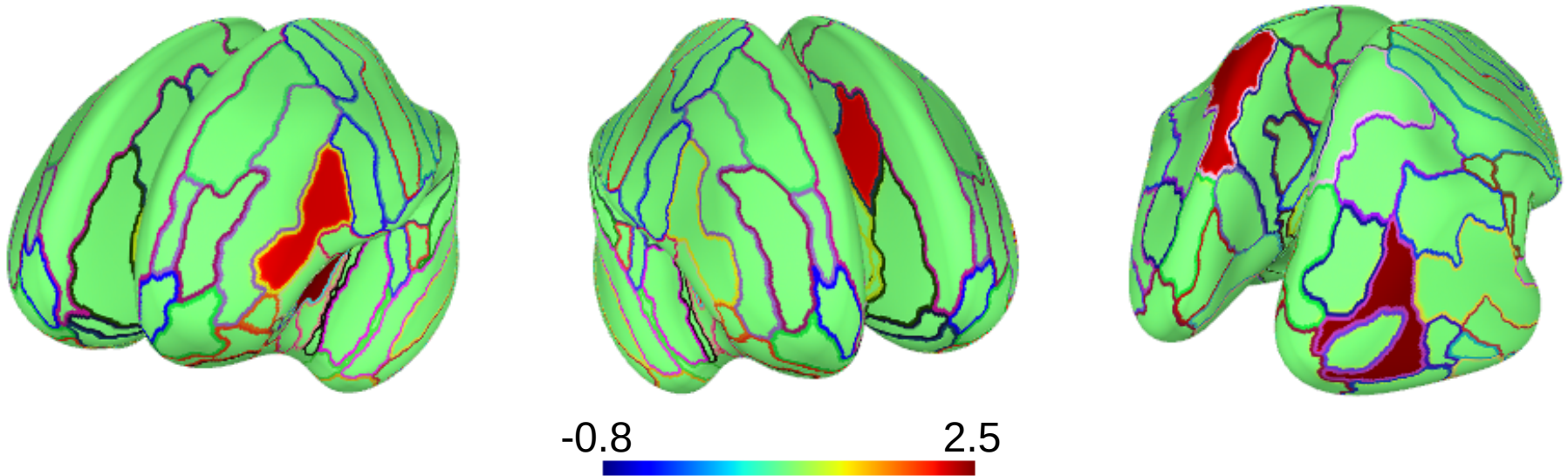
14 – 21 Hz, parm = group, img = “Human”, thresholded



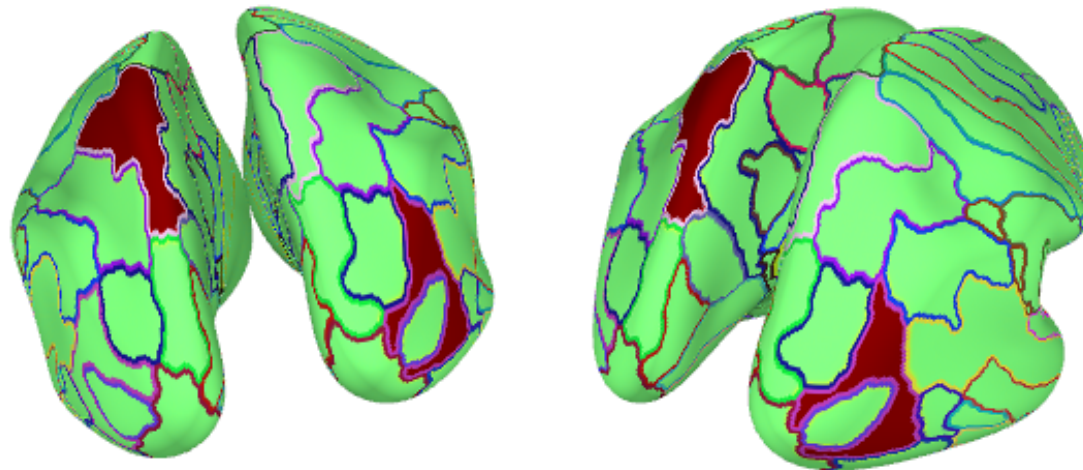
21 – 28 Hz, parm = group, img = “Super”, thresholded



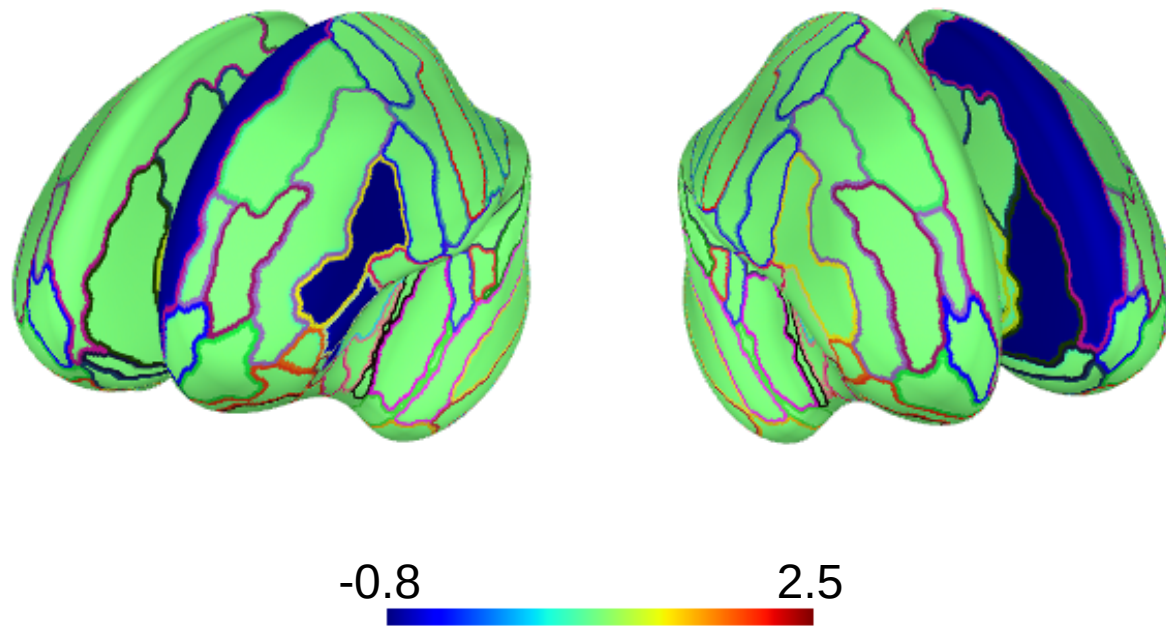
4 – 7 Hz, parm = religion, img = “God”, thresholded



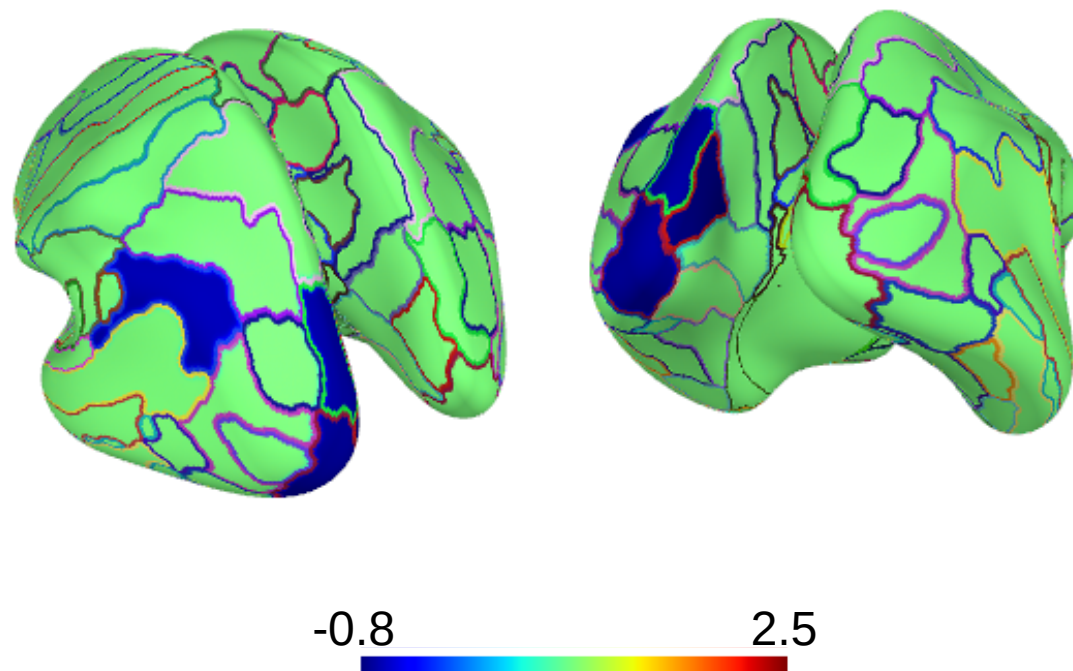
4 – 7 Hz, parm = religion, img = “Thing”, thresholded



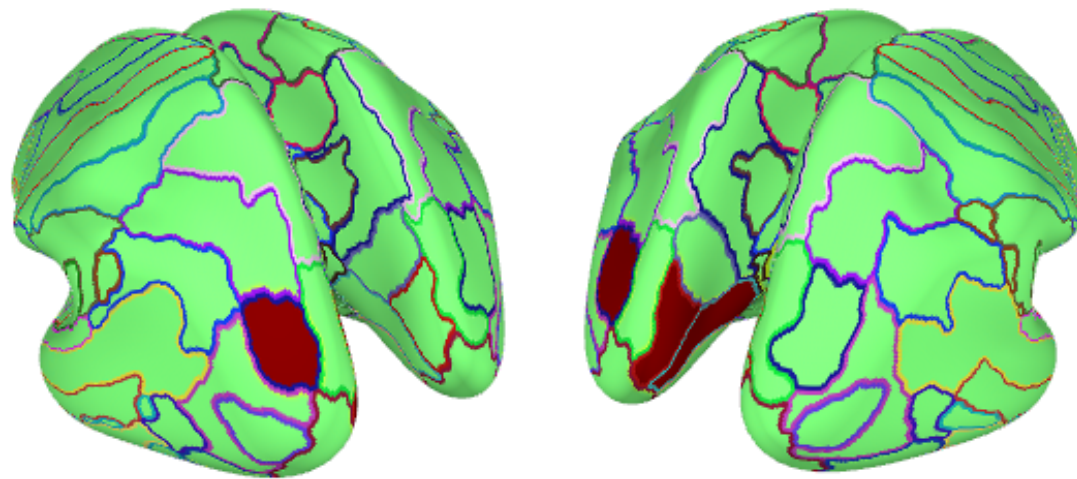
14 – 21 Hz, parm = religion, img = “Human”, thresholded



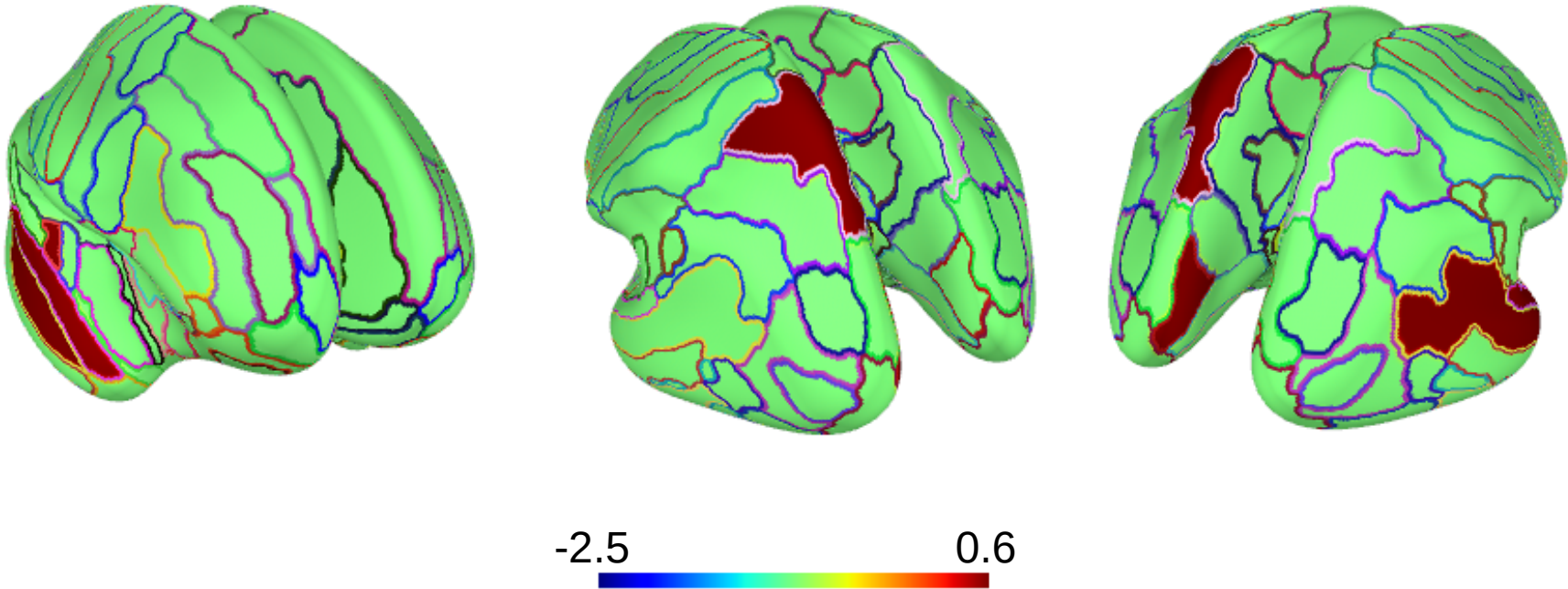
30 – 50 Hz, parm = religion, img = “Super”, thresholded



8 - 13 Hz, parm = group*religion, img = “God”, thresholded



8 - 13 Hz, parm = group*religion, img = “Super”, thresholded



30 - 50 Hz, parm = group*religion, img = “Human”, thresholded

